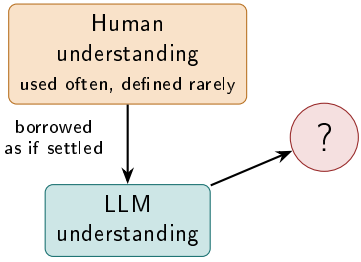
Is understanding an anthropocentric concept?

Soumya Banerjee

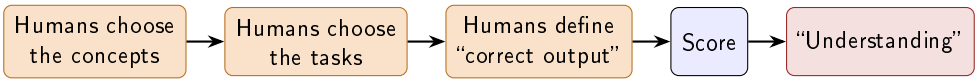
Computer Science, University of York, York, UK

The question behind the question

- Debates ask whether LLMs "understand" what they reason about.
- The question assumes one well-defined target that a system has or lacks.
- Yet understanding resists definition even in the human case: the term is used constantly and defined rarely.
- Changing to benchmark language obscures this difficulty rather than resolving it.

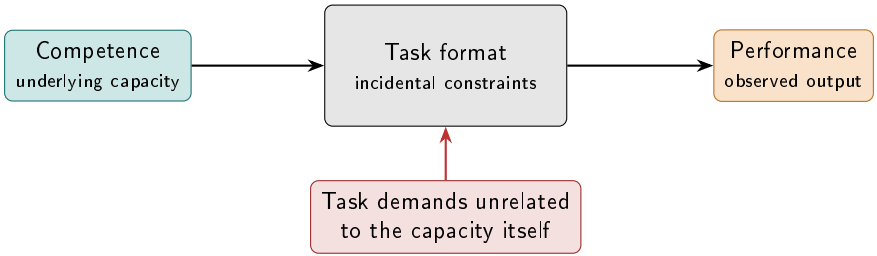


How the field measures understanding



- The evidence counted, the concepts checked, and the tasks that count as needing understanding are all human-chosen, calibrated for human cognitive architecture.
- Claim: the resulting picture is anthropocentric as a *structural feature*, not as an incidental bias better benchmarks could remove.

First problem: competence versus performance



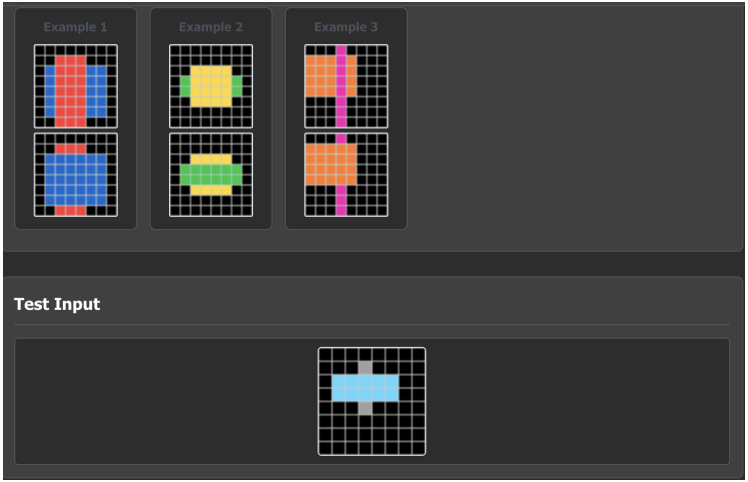
- A system may fail not because it lacks the capacity, but because the task format hides it (Firestone, 2020).
- A single scalar accuracy score conflates the two. ConceptARC, like most benchmarks, reports only the scalar.

The case study: ConceptARC



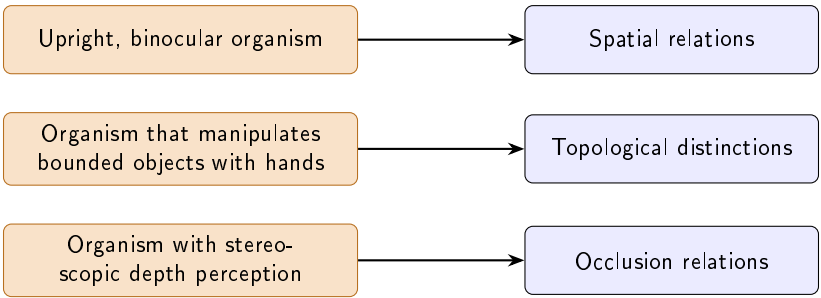
- The taxonomy was built to test whether systems generalise the way humans do.
- The human-versus-machine comparison is informative only if the taxonomy is a neutral coordinate system, not one native to a single party.

ConceptARC benchmark



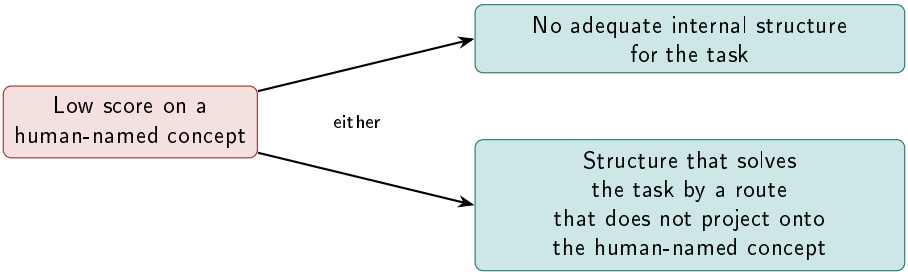
Example ConceptARC task: demonstration pairs and a test input.

Second problem: whose concepts are tested?



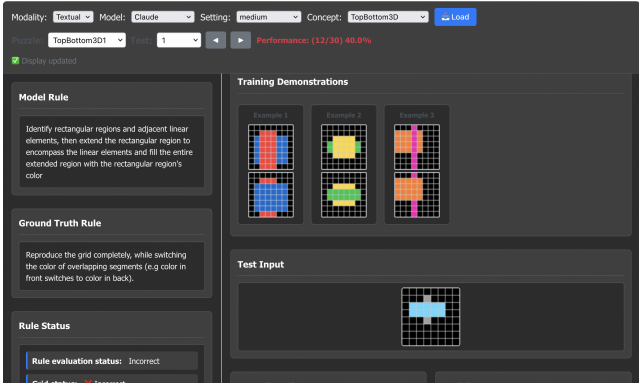
- The target concepts are carvings a human finds natural and easy to name. None is arbitrary for humans; all are chosen because they are natural for humans.

What does a low score mean?



- Read this way, the score measures **taxonomic alignment** more than understanding as such.

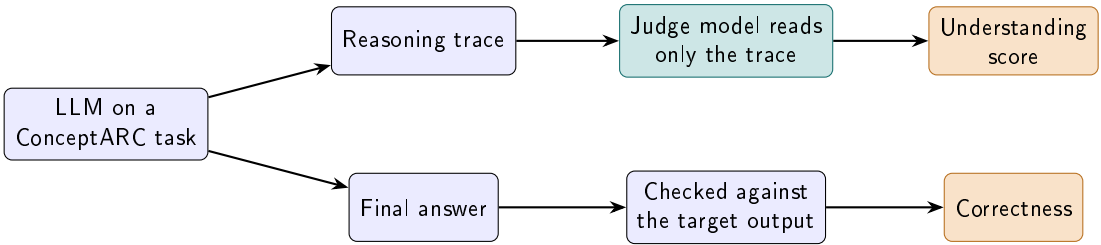
Machines may solve problems with different concepts



ConceptARC TopBottom3D puzzle: model rule versus ground-truth rule (Claude, textual modality, medium setting).

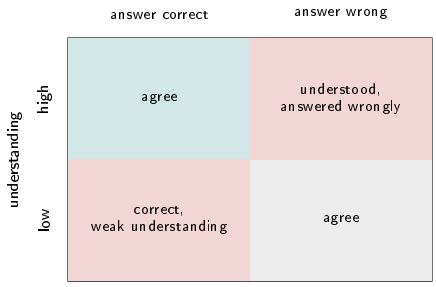
- **Ground truth (human concept)**: switch the colours of overlapping segments, front to back.
- **Model's rule**: find rectangular regions and adjacent linear elements, extend the rectangle, and fill it with its colour.
- The model describes the task in terms of geometry and fill, not layering.
- The rule is scored incorrect against the human-named concept. Missing structure, or a different route? That is the open question.

Separate understanding from correctness



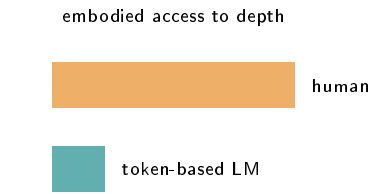
Finding 1: understanding and correctness dissociate

- The two measures come apart substantially *in both directions*.
- A correct final answer does not guarantee conceptual understanding, and conceptual understanding does not guarantee a correct answer.



Finding 2: one concept family is worst

- Lowest scores concentrate in a single family: concepts requiring reasoning about **depth, layering, and occlusion**.
- This is the family where human cognition has the most direct embodied purchase and a token-based language model has the least.



Qualitative schematic, not measured values.

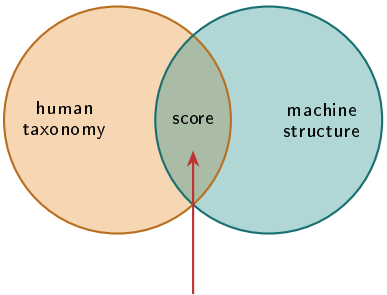
Three claims drawn from the pattern

1. "Concept understanding" as standardly measured is best read as alignment with a human-authored taxonomy, not a general measure of internal structure.

2. Where alignment breaks down tracks an asymmetry in embodied access to the physical world: a mismatch between task demands and the kind of system performing it.

3. The human case does not escape: the same labelling task given to human participants shows a structurally similar pattern of concept-specific confusion.

Understanding as taxonomic alignment



- The score reflects the overlap between the two.
- Machine structure outside the overlap is invisible to the test, though it may still support task success.
- Low overlap is evidence of misalignment first, and absence of competence only as a further hypothesis.

Related work: a vocabulary problem

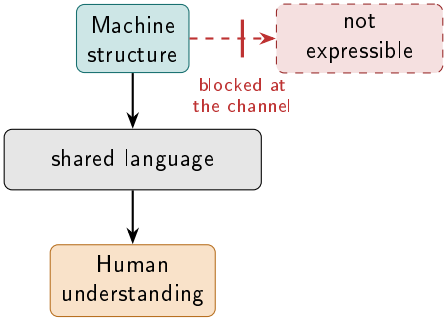
- Much vocabulary for machine cognition is imported from human psychology (Mitchell, 2024), and rich psychological terms need care in AI (Shevlin and Halina, 2019).
- A taxonomy built from that vocabulary defines the object of measurement in human terms from the start.
- Proposed response: treat interpretability as a communication problem and create new terms for machine-native concepts (Hewitt et al., 2025).



a shared vocabulary

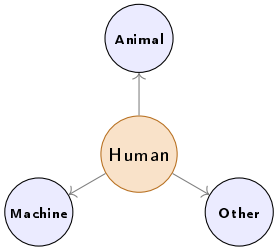
A prison-house of language?

- If understanding is a communication problem between machines and humans, every claim about machine understanding must pass through a language the two share.
- We may then always be confined to what that shared language can express.
- Machine-native structure outside the shared vocabulary stays invisible to us, as with the part of the machine circle outside the taxonomy overlap.
- Partial exits: new terms for machine-native concepts (Hewitt et al., 2025); channel-independent evidence such as representational probing.
- **Open question**: can any evidence escape human interpretation altogether?



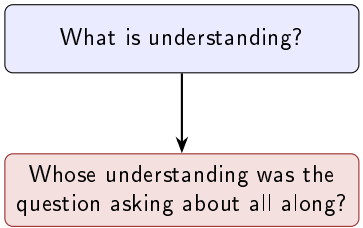
Related work: intelligence is task-relative

- Intelligence defined as an agent's capacity to achieve goals across a wide range of environments, silent on which ones (Legg and Hutter, 2007).
- A heliocentric rather than geocentric view: human cognition is one point among many, not the fixed centre (Holm and Banerjee, 2024).



Conclusion: whose understanding?

- No first-order theory of understanding is offered.
- The dominant measurement practice smuggles in an anthropocentric answer: understanding *is* alignment with a taxonomy native to embodied human cognition.
- Decentering the human case does not mean abandoning it: hold that vocabulary at arm's length, as one coordinate system among several.



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